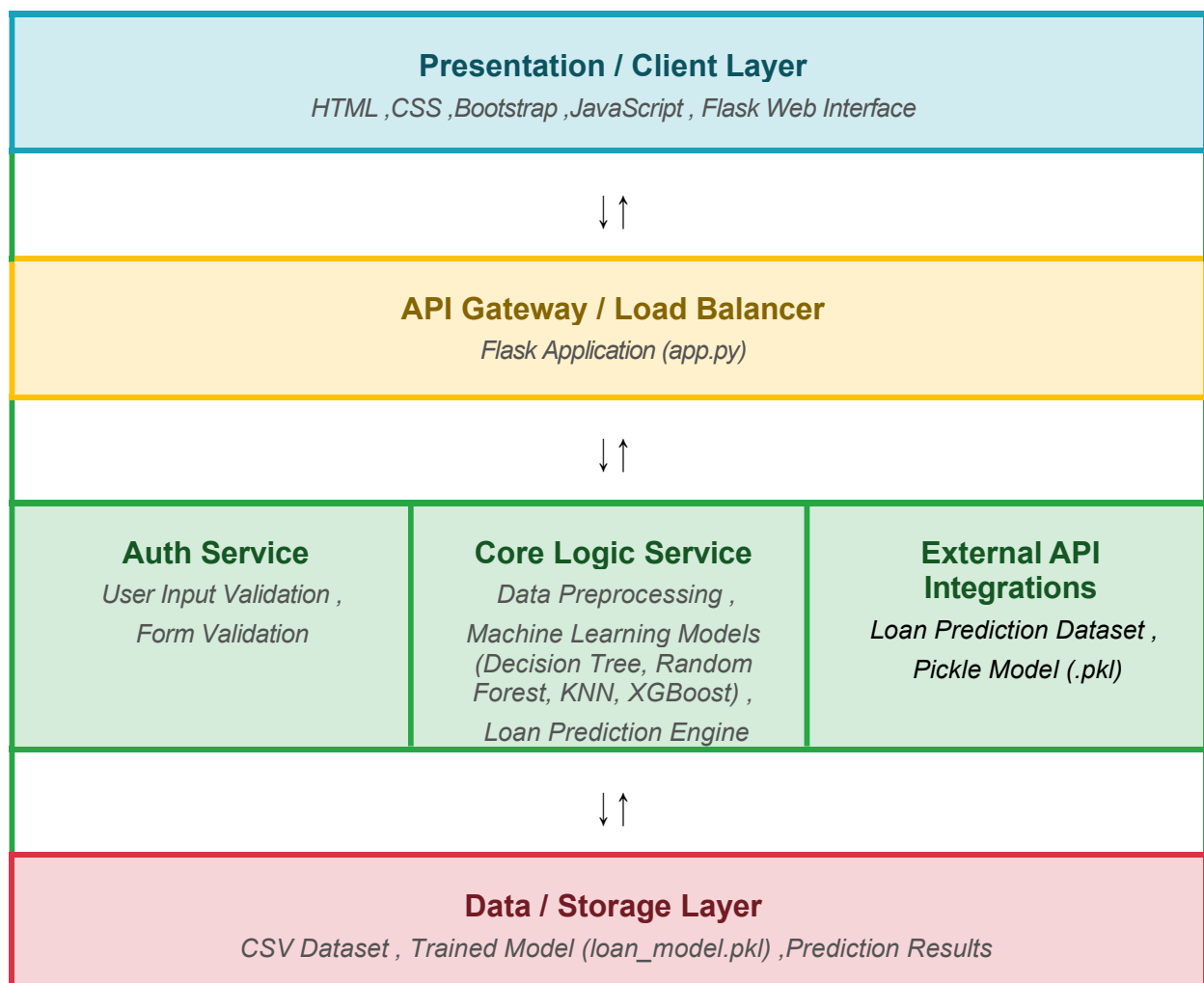


Solution Architecture

Date	25 May 2026
Team ID	XXXXXX
Project Name	Smart Lender – AI Loan Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface for entering applicant details and viewing loan prediction results.</i>	<i>HTML, CSS, Bootstrap, JavaScript</i>
<i>API Gateway / Backend</i>	<i>Handles user requests, validates inputs, and communicates with the prediction model.</i>	<i>Flask, Python</i>
<i>Core Logic Service</i>	<i>Preprocesses data, runs ML algorithms, and predicts loan approval status.</i>	<i>Scikit-learn, XGBoost, Pandas, NumPy</i>
<i>Data / Storage Layer</i>	<i>Stores the loan dataset, trained model, and prediction data.</i>	<i>CSV Dataset, Pickle (.pkl)</i>